QUEEN'S UNIVERSITY BELFAST

QUEEN'S BUSINESS SCHOOL

MGT7216

Correlation and Regression Analysis

Week 4

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Learning Objectives

- To consolidate the understanding of bivariate correlation analysis using various data types.
- To consolidate the understanding of simple and multiple linear regression analysis using various data types.
- To understand statistical tests like t, F, and Chi-square.
- To understand the basic model fit using regression analysis.
- To enhance the deeper understanding of the p-value.
- To understand how to infer insights from scatterplots and error distribution.

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Correlation Analysis

Correlation analysis is a statistical technique used to determine the strength and direction of the relationship between two or more variables.

It is a widely used technique in business to identify and quantify relationships between different business metrics.

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Business Context:

Human Resources: A company may want to determine the degree of relationship between employee satisfaction and employee productivity.

Marketing: A company may want to determine the degree of relationship between their marketing spend and their sales.

Operations: A company may want to determine the degree of relationship between the number of defects and the number of complaints.

Finance: A company may want to determine the degree of relationship between interest rates and stock prices.

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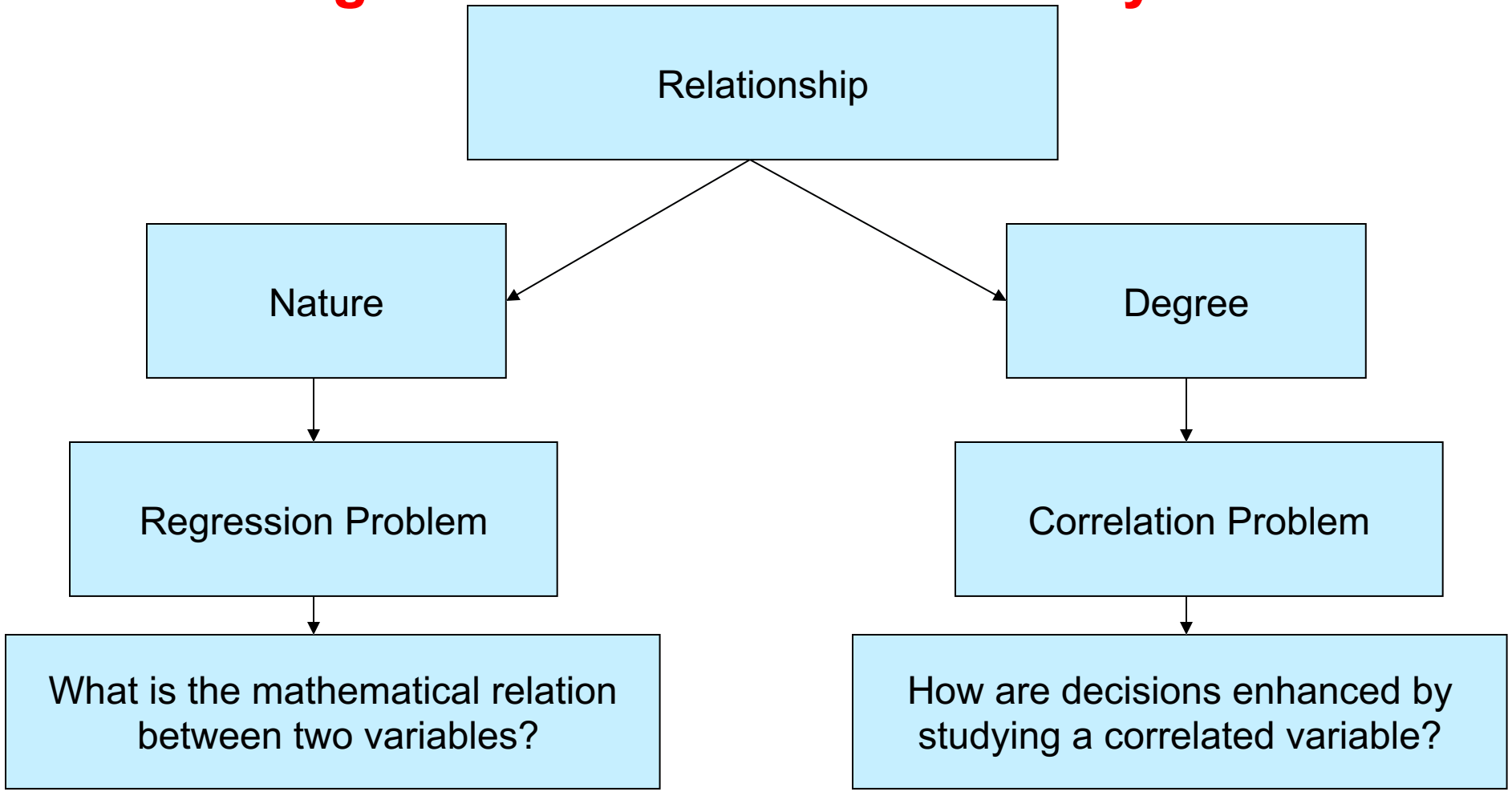
Schematic Overview of Regression & Correlation Analysis



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Correlation & Association

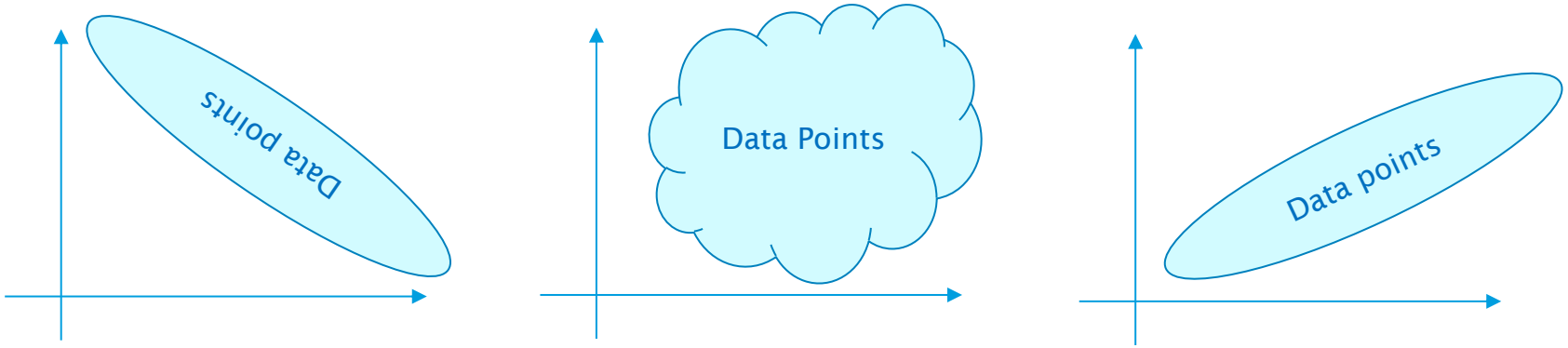
Categorical/ Nominal
Ordinal
Continuous

X ← r → Y

Categorical/ Nominal
Ordinal
Continuous

Simple correlation measures the strength of the relationship between two variables; also known as bivariate correlation.

Correlation coefficient r ranges between -1 and +1.



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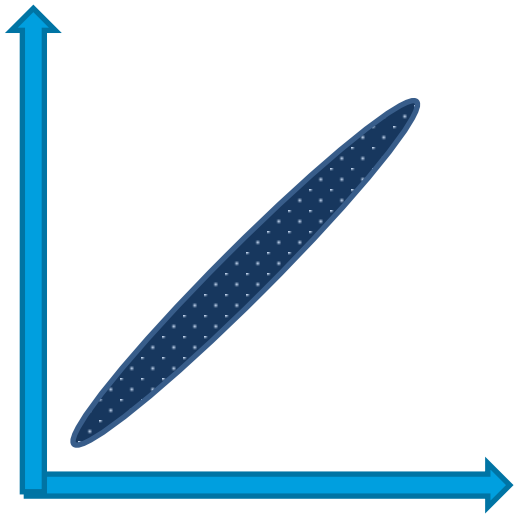
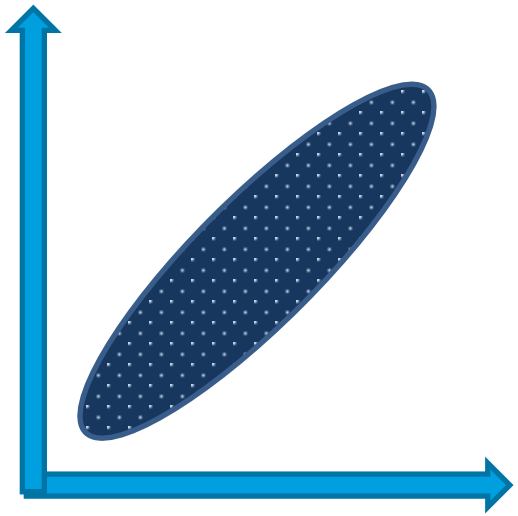
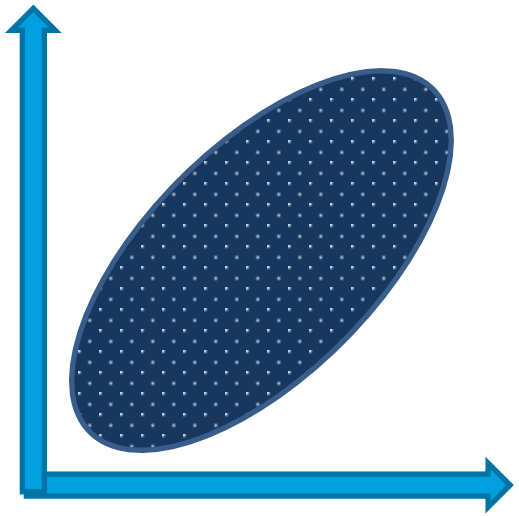
Scatter Plots



High positive correlation



Positive correlation



Low positive correlation

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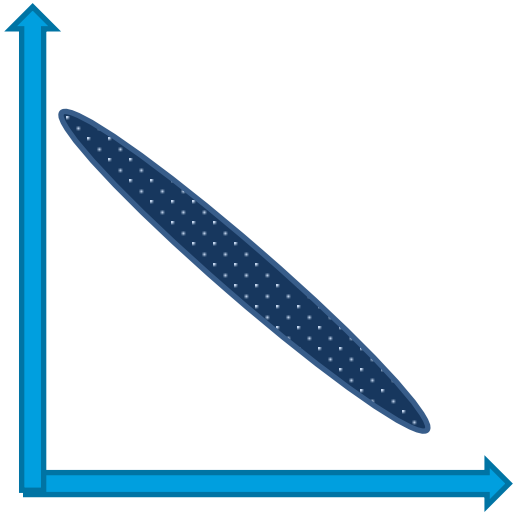
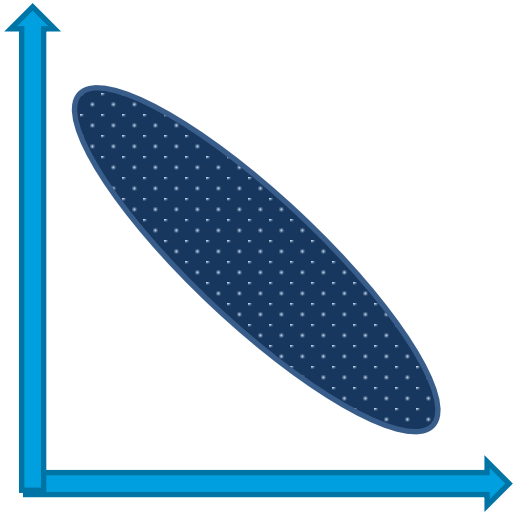
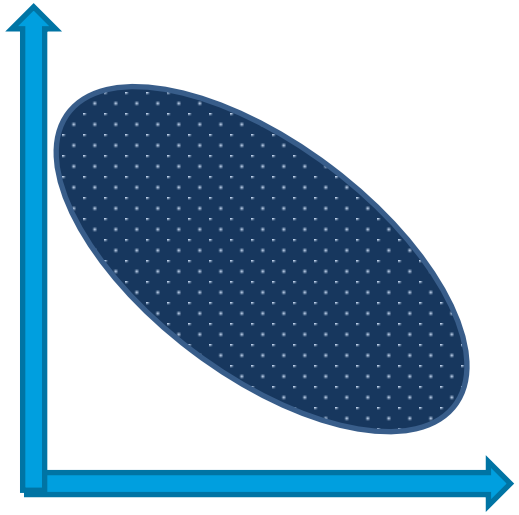
Scatter Plots



High negative correlation



Negative correlation



Low negative correlation

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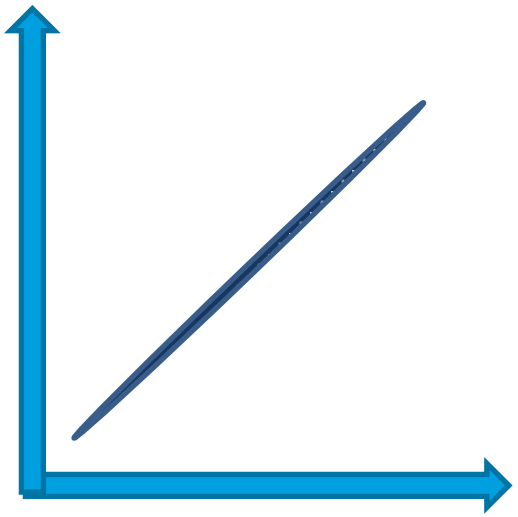
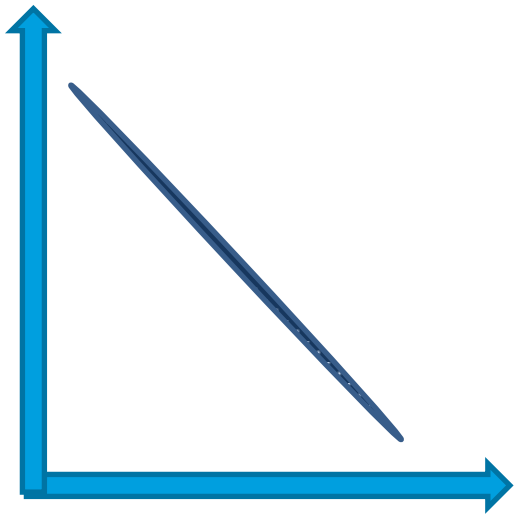
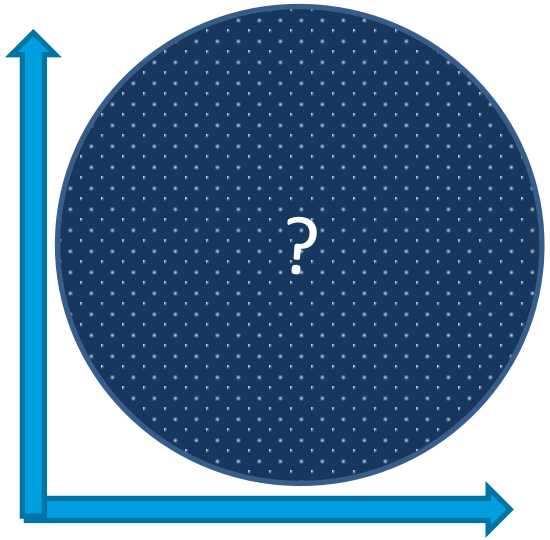
Scatter Plots



Perfect positive correlation



Perfect negative correlation



No Correlation ???

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Pearson correlation coefficient (r)

1 perfect positive correlation
.9 to 1 very high correlation
.7 to .9 high correlation
.5 to .7 moderate correlation
.3 to .5 low correlation
0 to .3 little if any correlation
0 no correlation

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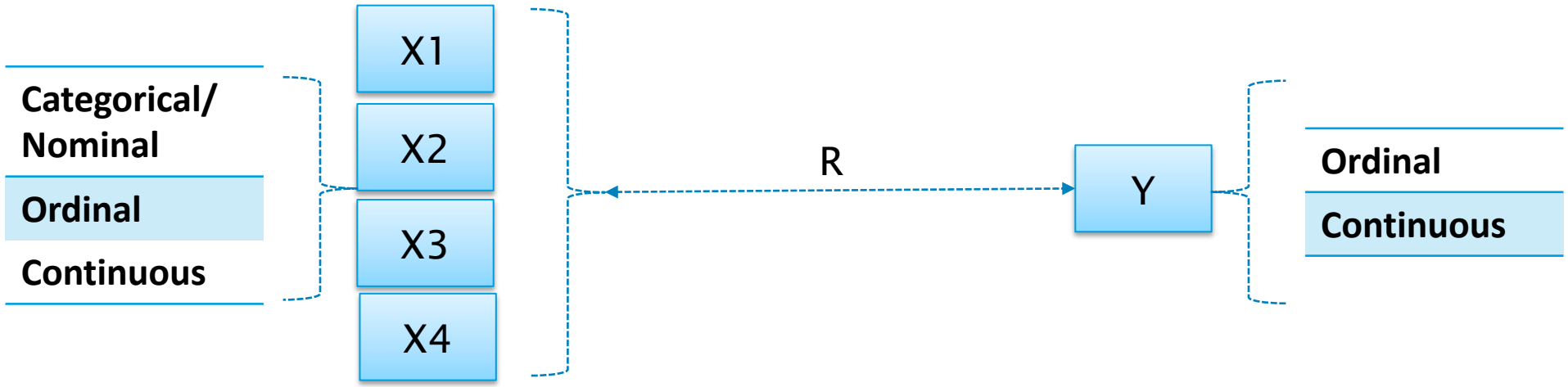
Multiple Correlation



Multiple correlation is used to determine the strength and direction of the linear relationship between a dependent variable and two or more independent variables.

Correlation coefficient R ranges between -1 and +1.

Business Context: The relationship between a company's revenue (dependent variable - Y) and several independent variables (Xs) such as advertising spend, number of employees, and product pricing.

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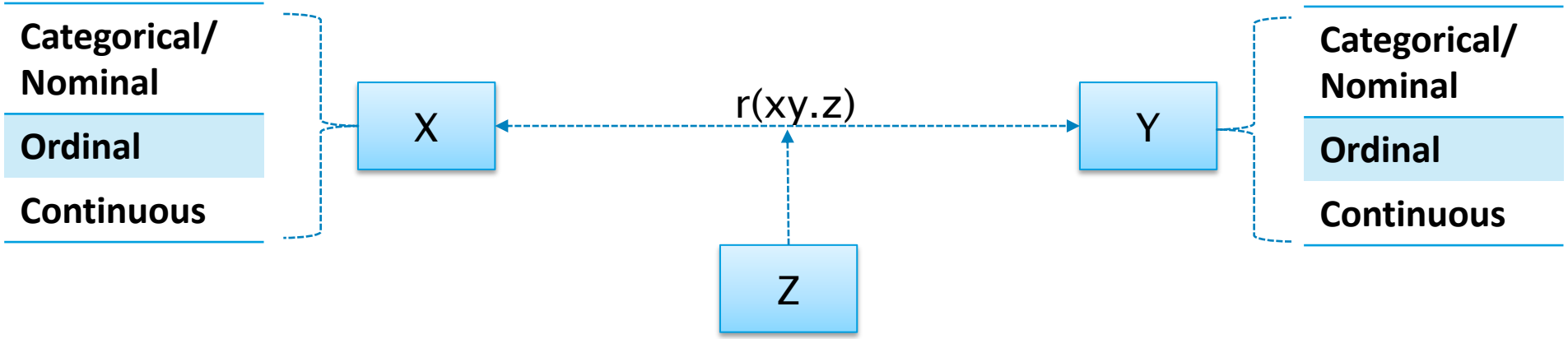
Partial Correlation



Partial correlation is used to determine the strength and direction of the linear relationship between two variables while controlling for the influence of one or more other variables.

Correlation coefficient r(xy.z) ranges between -1 and +1.

Business Context: Sales and Advertising: A business might want to understand the relationship between sales and advertising spend while controlling for the effect of other variables such as price and promotions.

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Correlations Methods

Type of Variables	Appropriate Correlation Method
Continuous vs Continuous	Pearson correlation
Ordinal vs Ordinal	Spearman correlation or Kendall correlation
Categorical vs Categorical	Cramer's V
Continuous vs Categorical (binary)	Point-biserial correlation
Continuous vs Categorical (non-binary)	Biserial correlation
Ordinal vs Categorical (binary)	Point-biserial correlation or Spearman correlation
Ordinal vs Categorical (non-binary)	Spearman correlation or Kendall correlation

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Correlations Methods

Variable Types	Appropriate Method	Business Case Examples
Continuous-Continuous	Pearson correlation	Temperature and ice cream sales
Categorical-Categorical (binary)	Point-biserial correlation or Phi coefficient	Gender and product preference
Categorical-Categorical (non-binary)	Cramer's V or Contingency Coefficient	Education level and occupation
Continuous-Ordinal	Spearman correlation or Kendall rank correlation	Age and satisfaction with a product on a Likert scale
Ordinal-Categorical (binary)	Point-biserial correlation	Job satisfaction (ordinal) and employee turnover (binary)
Ordinal-Categorical (non-binary)	Correlation ratio	Educational level (ordinal) and occupation (non-binary)

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Correlations Methods

Variable Types	Appropriate Method	Business Case Examples
Continuous-Categorical	Point-biserial correlation	Income (continuous) and job status (binary)
Ordinal-Ordinal	Spearman correlation or Kendall rank correlation	Income level (ordinal) and level of education (ordinal)

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Regression Analysis

Regression analysis is a statistical method used to determine the relationship between a dependent variable and one or more independent variables.

Regression analysis is important in business as it allows businesses to make predictions about future trends, identify the key drivers of business outcomes, and optimise decision-making.

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Business Context:

Human Resources: A company may want to determine the nature of relationship between employee satisfaction and employee productivity.

Marketing: A company may want to determine the nature of relationship between their marketing spend and their sales.

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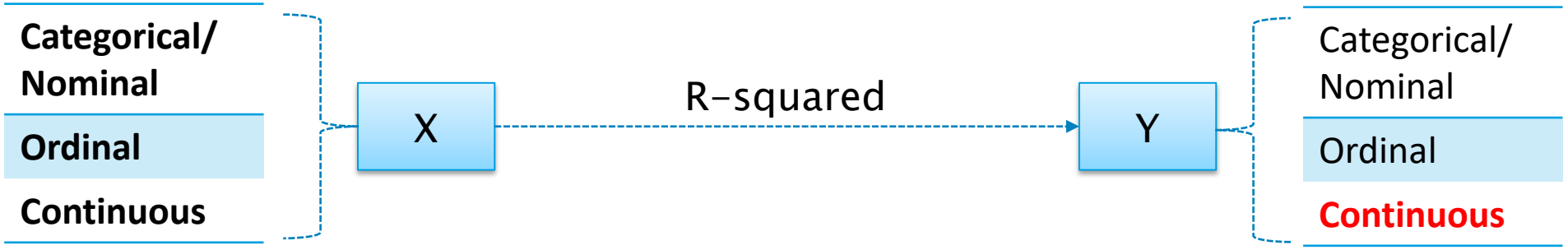
Linear Regression



Simple linear regression is a statistical method used to model the relationship between a dependent variable (Y) and a single independent variable (X).

Coefficient of determination R-squared ranges between 0 and +1, which determines the proportion of variance in the Y that can be explained by the X.

The assumptions of simple linear regression are linearity, independence, normality, and homoscedasticity.

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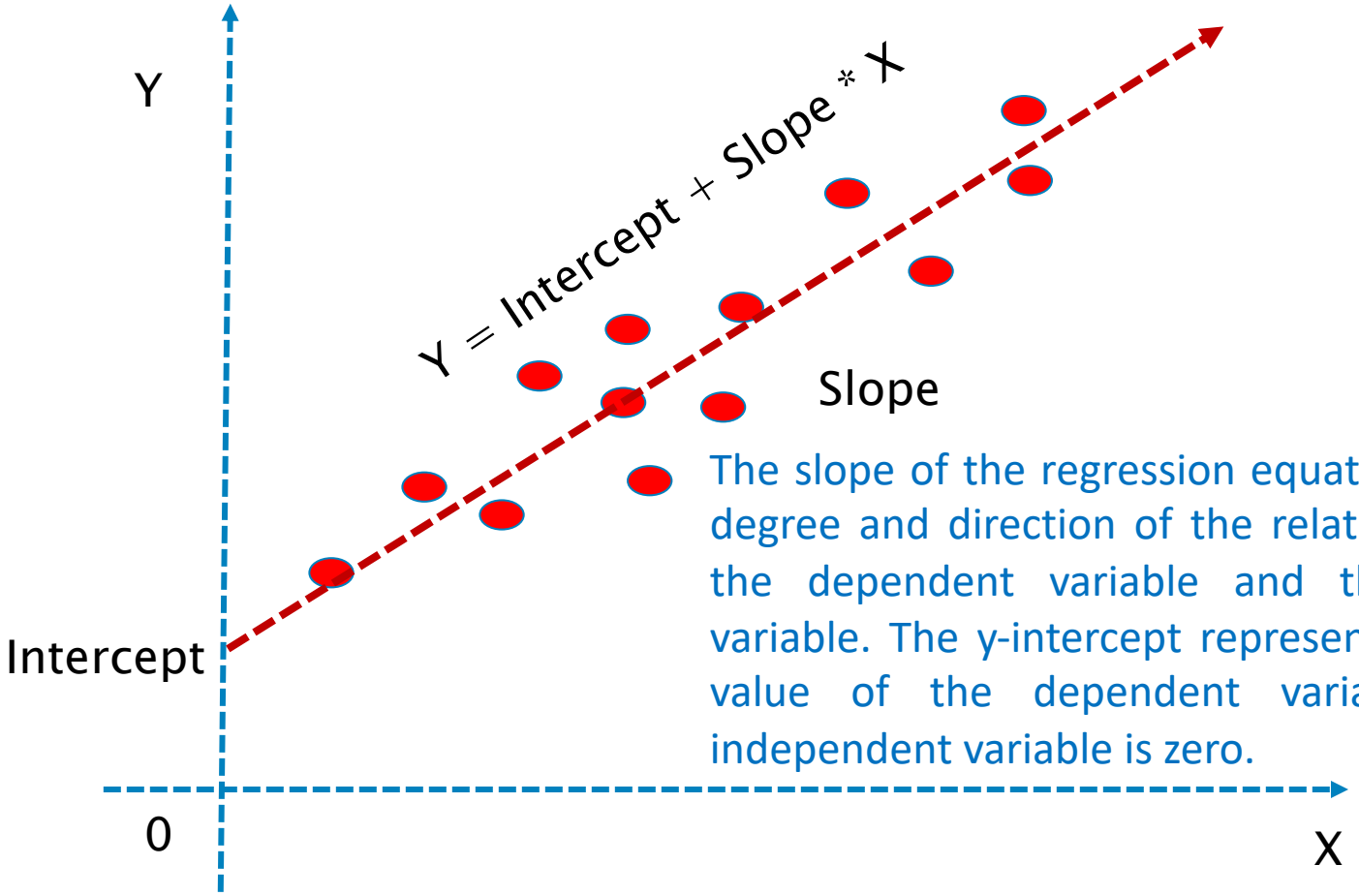
Regression Analysis



The slope of the regression equation indicates the degree and direction of the relationship between the dependent variable and the independent variable. The y-intercept represents the predicted value of the dependent variable when the independent variable is zero.

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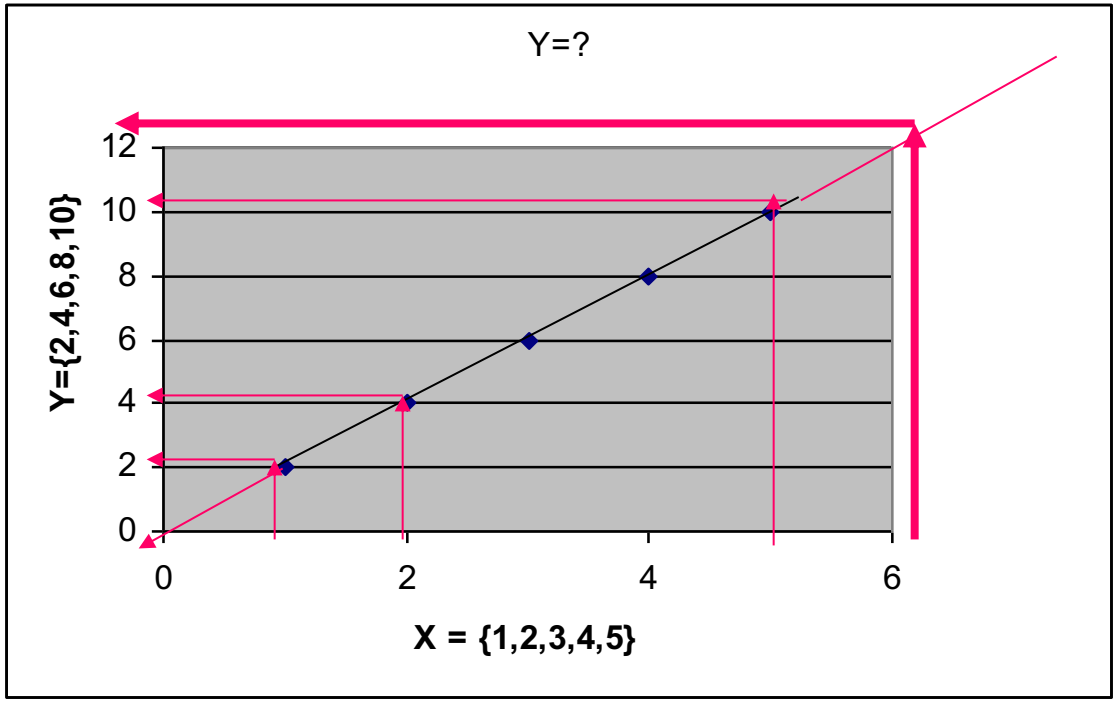
Regression Analysis



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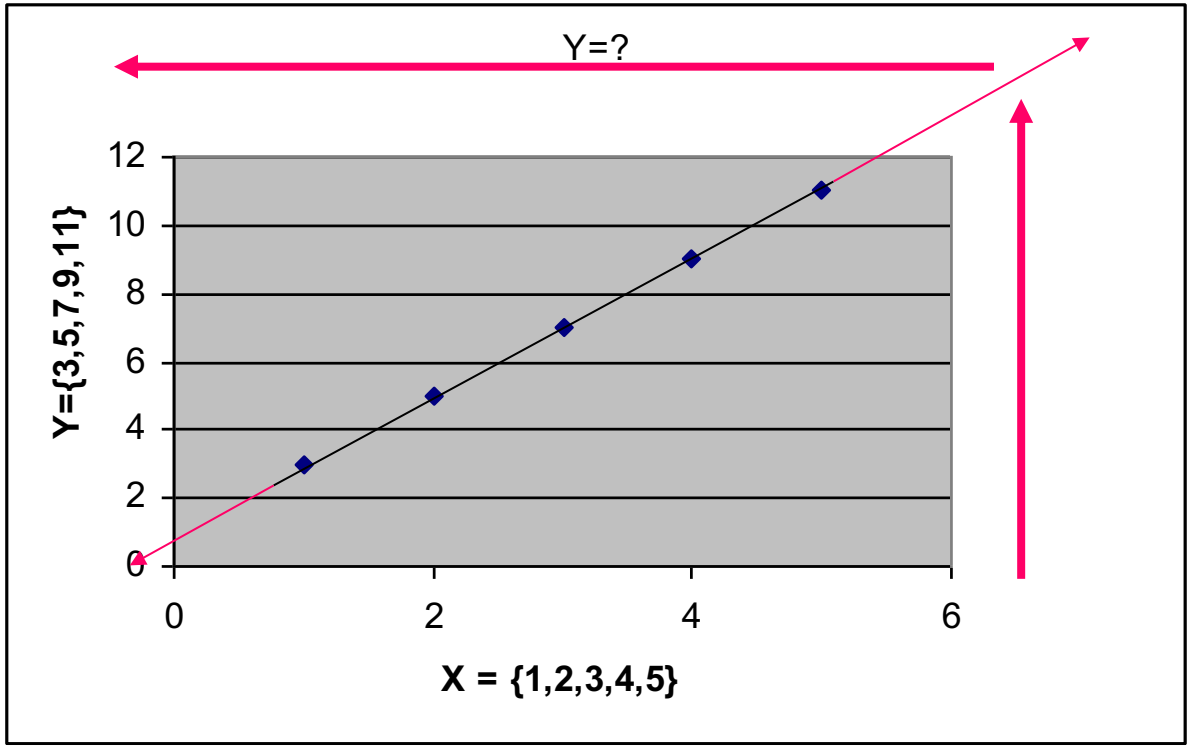
Regression Analysis



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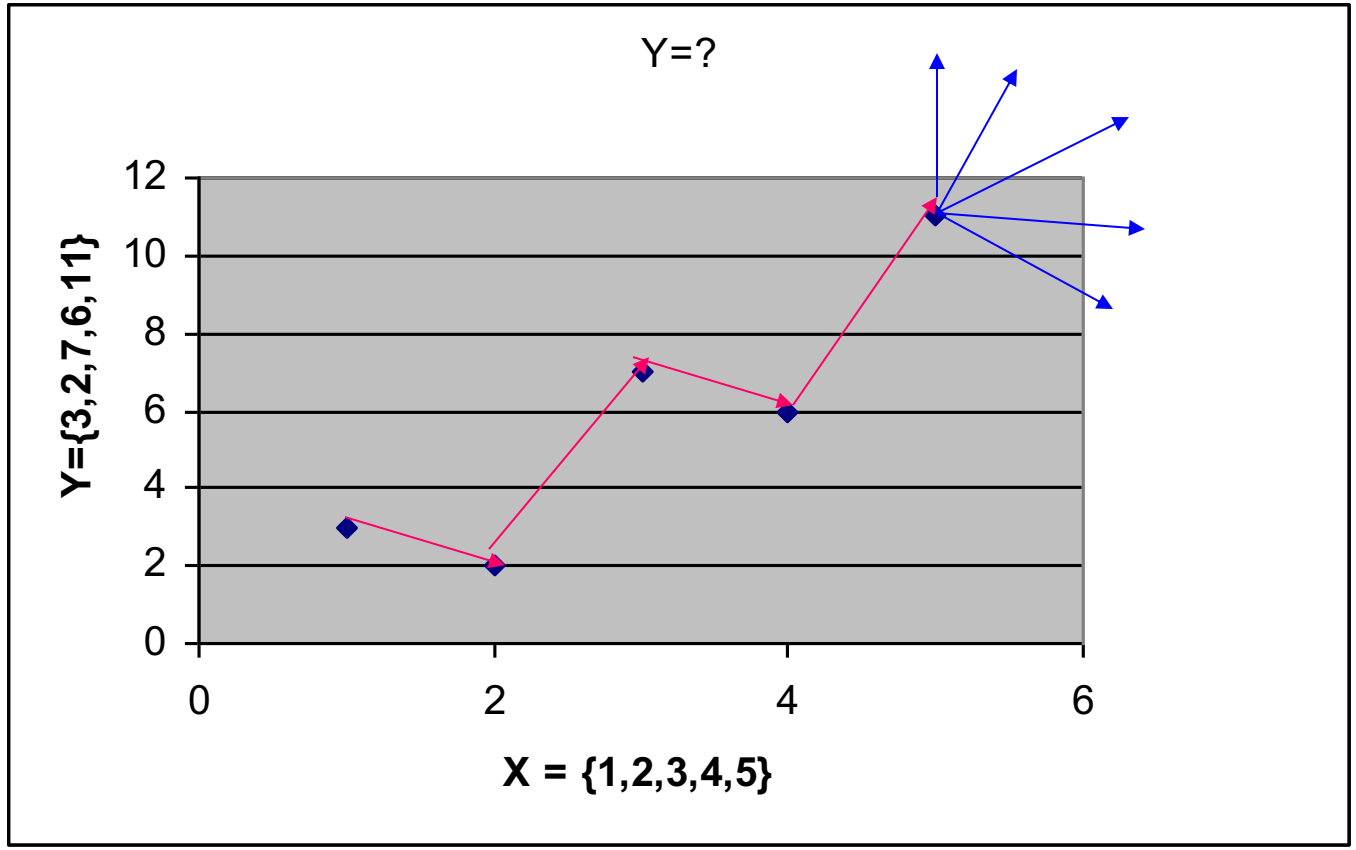
Regression Analysis



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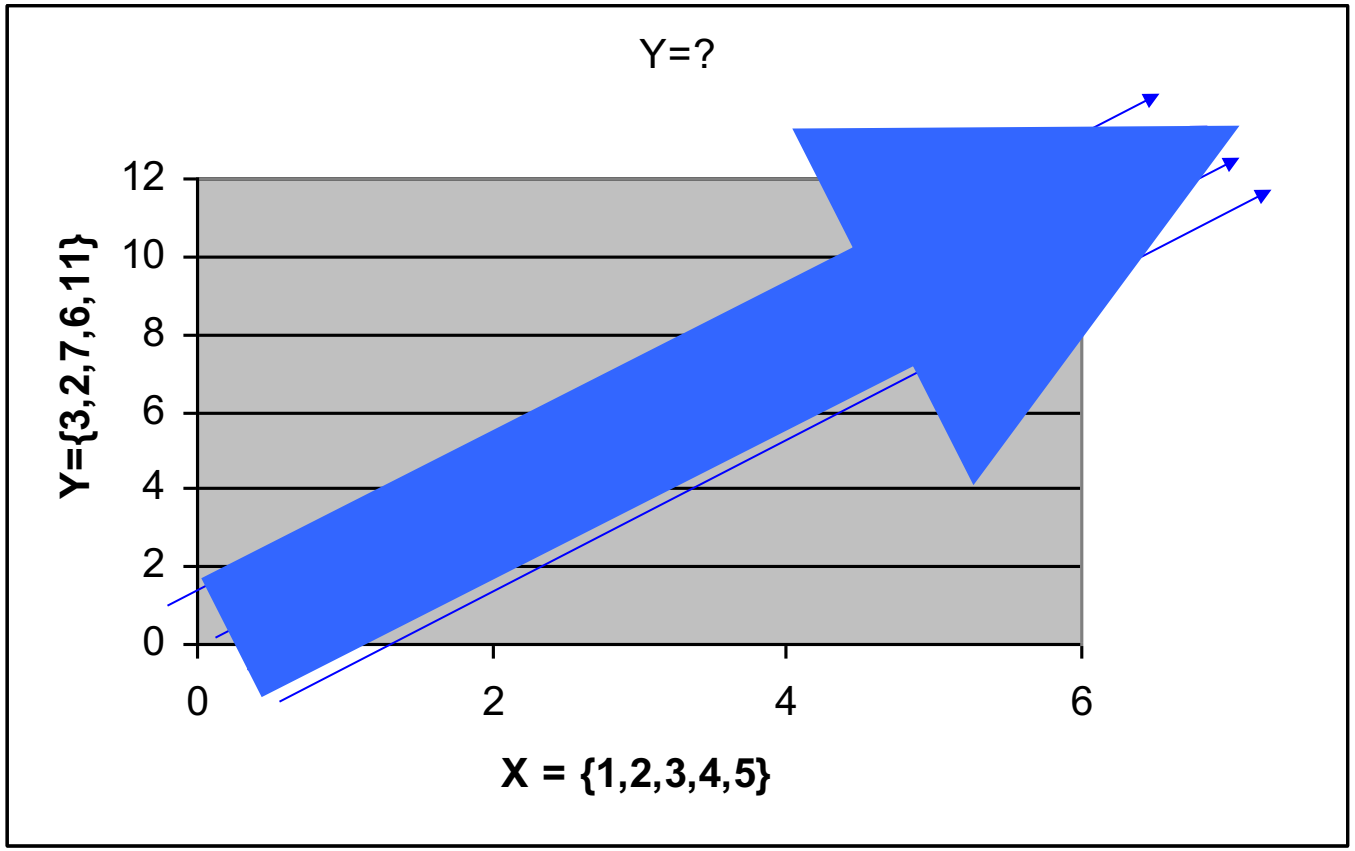
Regression Analysis



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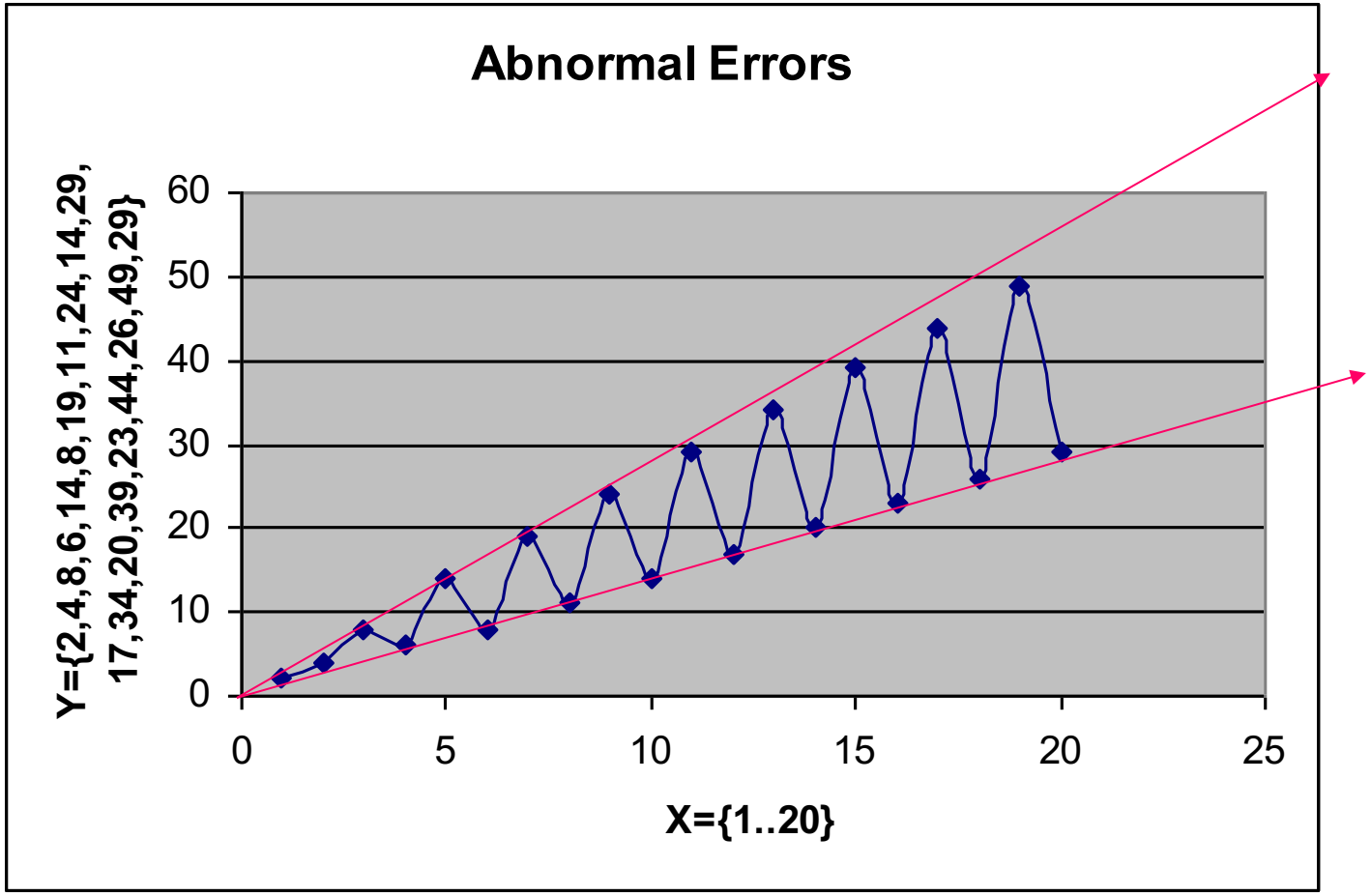

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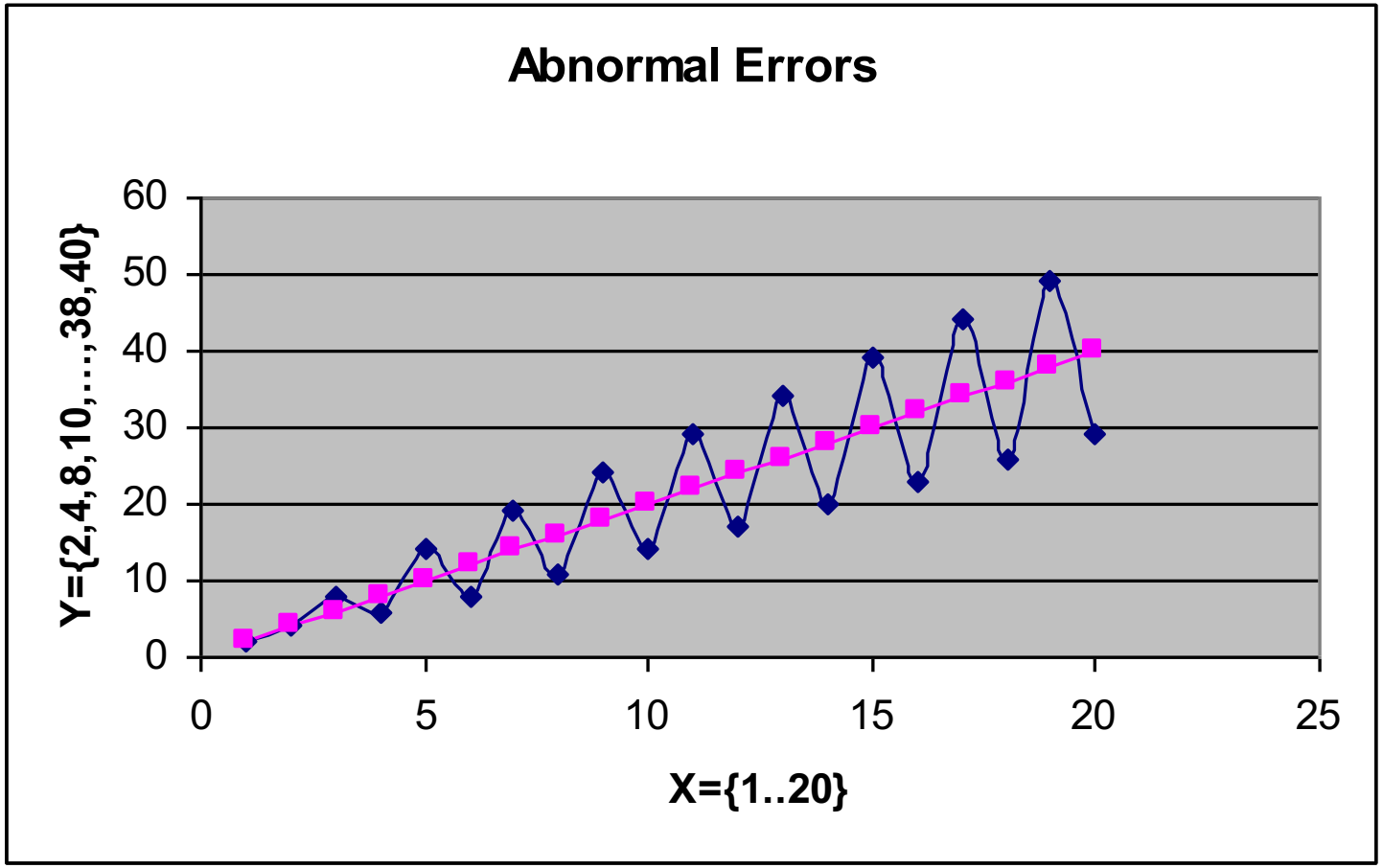
Regression Analysis



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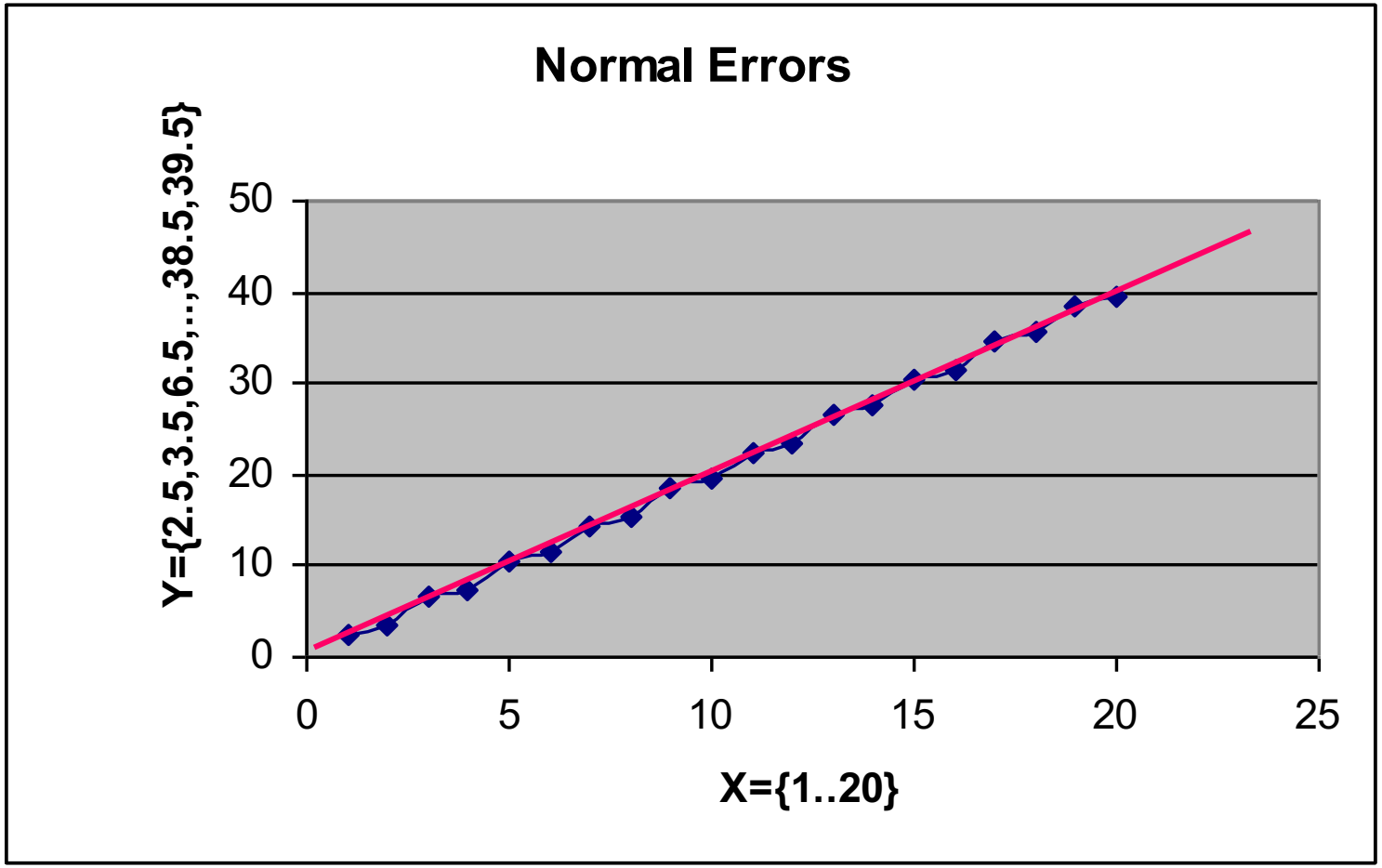
Regression Analysis



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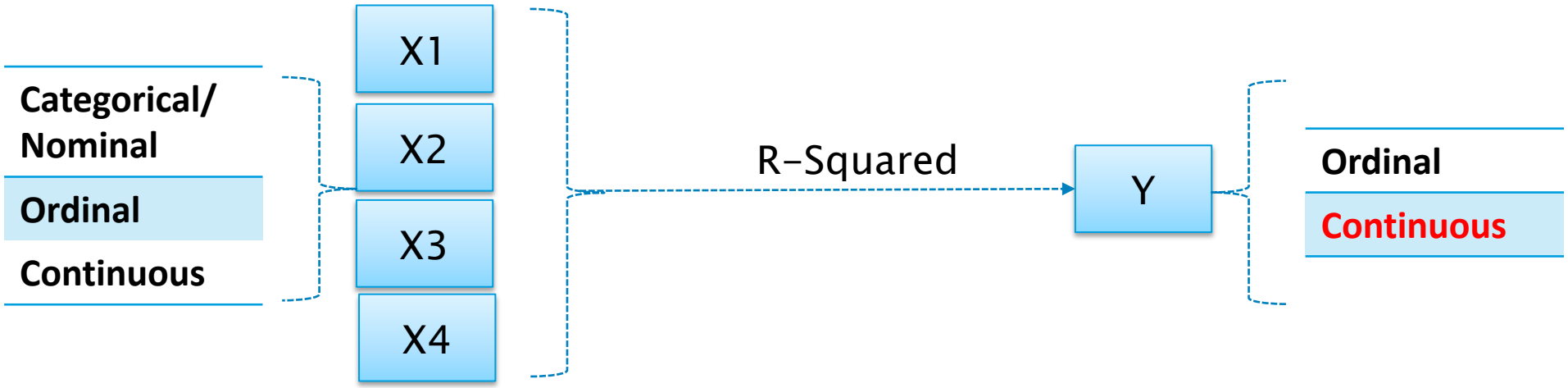
Multiple Linear Regression



Multiple linear regression is used to determine the nature of linear relationship between a dependent variable (Y) and two or more independent variables (Xs).

The assumptions of multiple linear regression are similar to those of simple linear regression.

Business Context: Predicting housing prices based on square footage, number of bedrooms and bathrooms, predicting customer satisfaction based on product quality, price and delivery time.

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Test Statistics

Test statistic	Purpose	Use case	Assumption	Strengths	Weaknesses
Pearson correlation coefficient (r)	To measure the degree of linear relationship between two variables	Continuous variables	Normal distribution of variables, Linearity	Easy to compute, can measure both positive and negative association	Only captures linear relationships
Spearman rank correlation coefficient (rho)	To measure the degree of monotonic relationship between two variables	Ordinal variables	Monotonic relationship	Robust to outliers, can capture nonlinear relationships	Less sensitive to small sample sizes, may not capture all types of nonlinear relationships

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Test Statistics

Test statistic	Purpose	Use case	Assumption	Strengths	Weaknesses
Kendall rank correlation coefficient (tau)	To measure the degree of association between two variables	Ordinal variables	Monotonic relationship	Robust to outliers, can capture nonlinear relationships	Less sensitive to small sample sizes, may not capture all types of nonlinear relationships
Chi-squared test	To determine if two categorical variables are independent	Categorical variables	Independence	Can handle large sample sizes, easy to interpret	Assumes large sample sizes, may have low power for small sample sizes

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Test Statistics

Test statistic	Purpose	Use case	Assumption	Strengths	Weaknesses
Simple linear regression t-test	To determine if there is a significant linear relationship between two continuous variables	Testing hypothesis about the slope parameter in a simple linear regression model	Linearity, constant variance, normality of residuals	Provides a measure of the strength and direction of the relationship, can test significance of the slope parameter	Assumes linearity and normality, sensitive to outliers
Multiple linear regression F-test	To determine if there is a significant linear relationship between two or more continuous variables and a dependent variable	Testing hypothesis about the significance of the model as a whole	Linearity, constant variance, normality of residuals, no multicollinearity	Can handle multiple predictors, can test significance of the model as a whole	Assumes linearity and normality, sensitive to outliers and multicollinearity

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P-value

In statistics, the *p*-value is a measure of the strength of evidence against the null hypothesis.

The null hypothesis is a statement that assumes that there is no relationship between the two variables being analysed.

It is important to note that a statistically significant result does not necessarily mean that the result is practically significant or important.

Additionally, a non-significant result does not necessarily mean that there is no relationship between the two variables; it may simply mean that the study was not designed or powered to detect such a relationship.

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Error Distribution in Regression Analysis

Error refers to the difference between the predicted value and the actual value of the dependent variable.

Understanding the error distribution in regression is important for evaluating the goodness of fit of the model and making accurate predictions.

A normal distribution with constant variance is an ideal assumption, but it may not always hold in practice.

It is important to identify the appropriate distribution for the data and adjust the model accordingly to obtain reliable results.

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Error Distribution in Regression Analysis

In machine learning, one commonly used measure of error is the mean squared error (MSE), which is calculated as the average of the squared difference between the predicted and actual outcomes.

To minimise the MSE, machine learning algorithms use various optimisation techniques such as gradient descent, which updates the model parameters iteratively to minimise the error.

Although machine learning algorithms do not typically assume a specific distribution for the errors, understanding the error distribution can still be useful for interpreting and validating the model results.

Other measures of errors: RMSE – Root mean squared error and MAE – Mean absolute error.

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Summary

- Correlation test => Measures the strength of correlation between two continuous variables.
- Using regression analysis, we establish causation between a DV and IV(s).
- Simple Linear Regression => Measures the strength of a one-directional relation between a DV and an IV.
- Multiple Linear Regression => Measures the strength of a one-directional relation between a DV and more than one IV.
- Make sure that your regression models meet data assumptions.

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References

1. Charles, V., & Gherman, T. (2025). Artificial Intelligence for Business. Sage. [Book in preparation].
2. Charles, V., & Emrouznejad, A. (Eds.) (2025). Artificial Intelligence for Business: State-of- the-art in prescriptive analytics research. Springer. [Edited book in preparation].

Thank you
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Questions

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